THE BASE CHANGE COMPARISON MAP IS AN ISOMORPHISM UNDER TOR INDEPENDENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Tor independence of two algebras over a ring. defined immediately before the statement, source lines 14945-14952. Supplies the hypothesis that makes the tensored resolution a resolution.
- The comparison map whose source and target are the two base changed complexes. constructed earlier in the same section, source lines 14895-14953. Supplies the map whose being an isomorphism is asserted.
- Derived tensor product computed by total complexes against free resolutions. used at proof lines 14962-14973 in the displayed chain of isomorphisms. Performs the four identifications that constitute the proof.

Source reference. Original lemma 0661: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7920–7920. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 065Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7918–7918. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *The comparison map (065Z) is an isomorphism if $A' = A \otimes_R R'$ and A and R' are Tor independent over R.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. To prove this we choose a free resolution $F^\bullet \to R'$ of R' as an R-module. Because A and R' are Tor independent over R we see that $F^\bullet \otimes_R A$ is a free A-module resolution of A' over A. By our general construction of the derived tensor product above we see that

$$K^\bullet \otimes_A A' \cong \mathrm{Tot}(K^\bullet \otimes_A (F^\bullet \otimes_R A)) = \mathrm{Tot}(K^\bullet \otimes_R F^\bullet) \cong \mathrm{Tot}(E^\bullet \otimes_R F^\bullet) \cong E^\bullet \otimes_R R'$$

as desired. □